Assignment 5C. Laguerre's method Marks 20

Posted on 09.09.2026 @ 2:30 pm and due on 09.09.2026 @ 5:40 pm

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1. Write the library functions for Laguarre's method and deflation for real roots only. Pass only the coefficients to the subroutines. **[5+5]**

2. Use Laguerre's method and deflation to find the roots (all real) of the following three polynomials. **[3+3+4]**

$$P_1(x) = x^4 - x^3 - 7x^2 + x + 6$$
$$P_2(x) = x^4 - 5x^2 + 4 = 0$$
$$P_3(x) = 2x^5 - 19.5x^3 + 0.5x^2 + 13.5x - 4.5$$